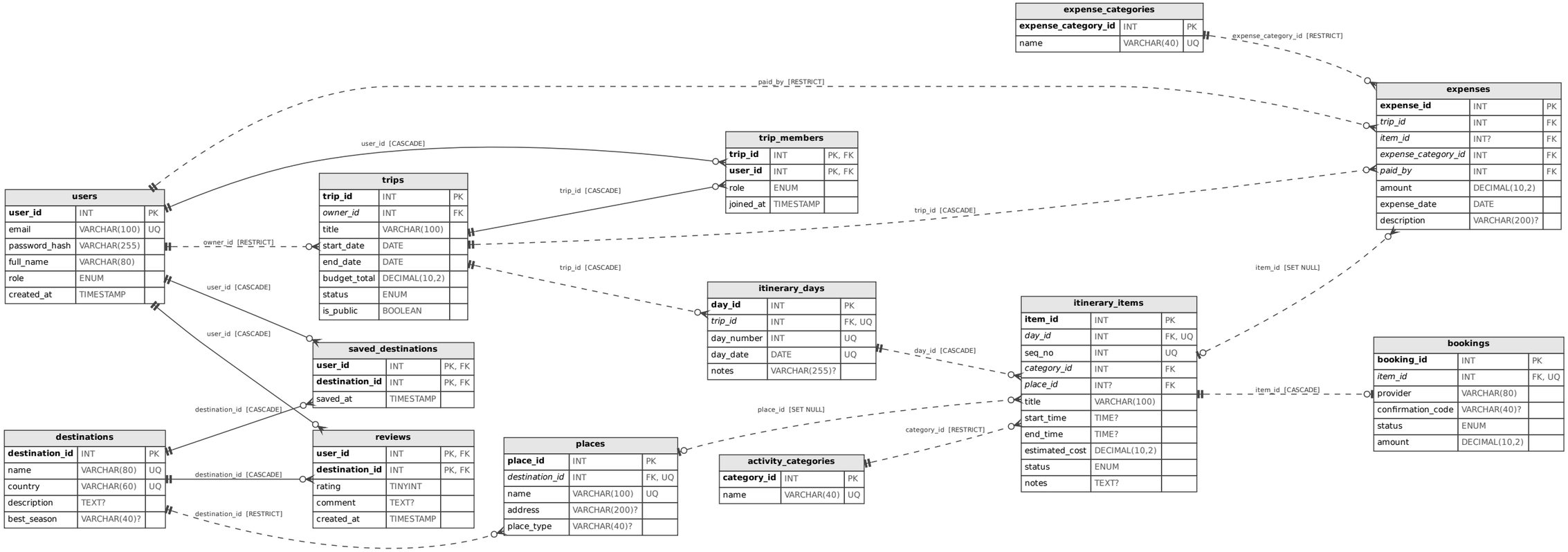
Full ER diagram: all 32 tables and 41 foreign keys



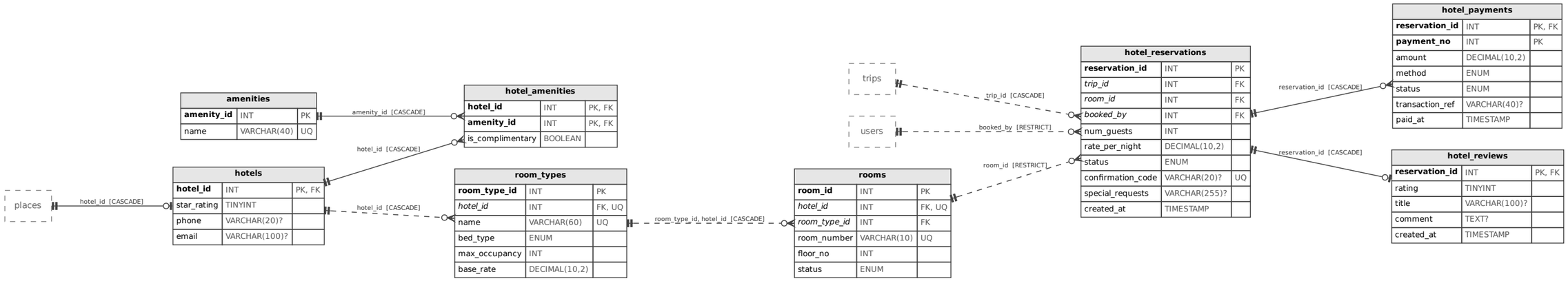
PK bold | FK italic | UQ unique | ? nullable | two bars = exactly one | bar and circle = zero or one | crow's foot and circle = zero or many | solid line = identifying (FK is part of the child's key), dashed = non-identifying | label = ON DELETE rule (every foreign key is ON UPDATE CASCADE)

Core module: trips, itinerary, expenses, community



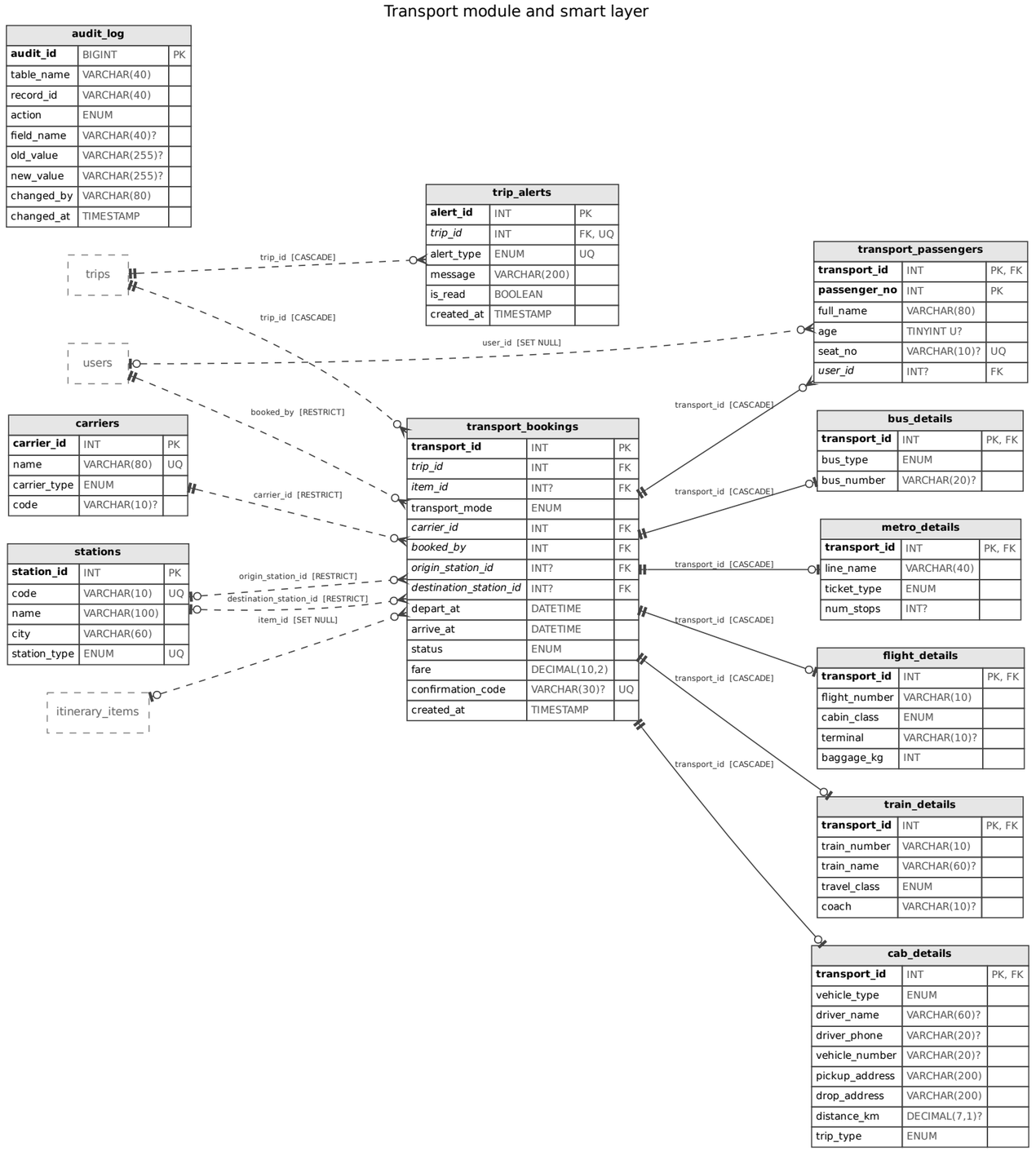
PK bold | FK italic | UQ unique | ? nullable | two bars = exactly one | bar and circle = zero or one | crow's foot and circle = zero or many | solid line = identifying (FK is part of the child's key), dashed = non-identifying | label = ON DELETE rule (every foreign key is ON UPDATE CASCADE)

Hotel module



PK bold | FK italic | UQ unique | ? nullable | two bars = exactly one | bar and circle = zero or one | crow's foot and circle = zero or many | solid line = identifying (FK is part of the child's key), dashed = non-identifying | label = ON DELETE rule (every foreign key is ON UPDATE CASCADE)

Transport module and smart layer



PK bold | FK italic | UQ unique | ? nullable | two bars = exactly one | bar and circle = zero or one | crow's foot and circle = zero or many | solid line = identifying (FK is part of the child's key), dashed = non-identifying | label = ON DELETE rule (every foreign key is ON UPDATE CASCADE)

Notation and all 41 relationships

Reading the diagrams: a table box lists every column with its type. Bold = primary key, italic = foreign key, UQ = part of a unique key, a trailing question mark = the column allows NULL. At each end of a line the symbol shows the cardinality: two bars mean exactly one, a bar with a circle means zero or one, a crow's foot with a circle means zero or many. A solid line is an identifying relationship (the foreign key is part of the child's primary key, as in weak entities and IS-A subtypes); a dashed line is non-identifying. Each line is labelled with its ON DELETE rule: CASCADE removes dependent rows, RESTRICT blocks the delete, SET NULL keeps the row and clears the link. Every foreign key is also ON UPDATE CASCADE.

Module pages: page 2 core, page 3 hotel module, page 4 transport module and smart layer. Dashed grey boxes on a module page are tables from another module, shown only as link targets. Two IS-A patterns: **places to hotels** and **transport_bookings to its five detail tables**. Weak entities: **hotel_payments, hotel_reviews, transport_passengers**. Many-to-many junctions: **trip_members, hotel_amenities, reviews, saved_destinations**.

#	Child table	FK column(s)	Parent table	Parent : child	Line	ON DELETE
1	places	destination_id	destinations	1 : 0..N	dashed	RESTRICT
2	trips	owner_id	users	1 : 0..N	dashed	RESTRICT
3	trip_members	trip_id	trips	1 : 0..N	solid	CASCADE
4	trip_members	user_id	users	1 : 0..N	solid	CASCADE
5	itinerary_days	trip_id	trips	1 : 0..N	dashed	CASCADE
6	itinerary_items	day_id	itinerary_days	1 : 0..N	dashed	CASCADE
7	itinerary_items	category_id	activity_categories	1 : 0..N	dashed	RESTRICT
8	itinerary_items	place_id	places	0..1 : 0..N	dashed	SET NULL
9	bookings	item_id	itinerary_items	1 : 0..1	dashed	CASCADE
10	expenses	trip_id	trips	1 : 0..N	dashed	CASCADE
11	expenses	item_id	itinerary_items	0..1 : 0..N	dashed	SET NULL
12	expenses	expense_category_id	expense_categories	1 : 0..N	dashed	RESTRICT
13	expenses	paid_by	users	1 : 0..N	dashed	RESTRICT
14	reviews	user_id	users	1 : 0..N	solid	CASCADE
15	reviews	destination_id	destinations	1 : 0..N	solid	CASCADE
16	saved_destinations	user_id	users	1 : 0..N	solid	CASCADE
17	saved_destinations	destination_id	destinations	1 : 0..N	solid	CASCADE
18	hotels	hotel_id	places	1 : 0..1	solid	CASCADE
19	hotel_amenities	hotel_id	hotels	1 : 0..N	solid	CASCADE
20	hotel_amenities	amenity_id	amenities	1 : 0..N	solid	CASCADE
21	room_types	hotel_id	hotels	1 : 0..N	dashed	CASCADE
22	rooms	room_type_id, hotel_id	room_types	1 : 0..N	dashed	CASCADE
23	hotel_reservations	trip_id	trips	1 : 0..N	dashed	CASCADE
24	hotel_reservations	room_id	rooms	1 : 0..N	dashed	RESTRICT
25	hotel_reservations	booked_by	users	1 : 0..N	dashed	RESTRICT
26	hotel_payments	reservation_id	hotel_reservations	1 : 0..N	solid	CASCADE
27	hotel_reviews	reservation_id	hotel_reservations	1 : 0..1	solid	CASCADE

#	Child table	FK column(s)	Parent table	Parent : child	Line	ON DELETE
28	transport_bookings	trip_id	trips	1 : 0..N	dashed	CASCADE
29	transport_bookings	item_id	itinerary_items	0..1 : 0..N	dashed	SET NULL
30	transport_bookings	carrier_id	carriers	1 : 0..N	dashed	RESTRICT
31	transport_bookings	booked_by	users	1 : 0..N	dashed	RESTRICT
32	transport_bookings	origin_station_id	stations	0..1 : 0..N	dashed	RESTRICT
33	transport_bookings	destination_station_id	stations	0..1 : 0..N	dashed	RESTRICT
34	flight_details	transport_id	transport_bookings	1 : 0..1	solid	CASCADE
35	train_details	transport_id	transport_bookings	1 : 0..1	solid	CASCADE
36	cab_details	transport_id	transport_bookings	1 : 0..1	solid	CASCADE
37	bus_details	transport_id	transport_bookings	1 : 0..1	solid	CASCADE
38	metro_details	transport_id	transport_bookings	1 : 0..1	solid	CASCADE
39	transport_passengers	transport_id	transport_bookings	1 : 0..N	solid	CASCADE
40	transport_passengers	user_id	users	0..1 : 0..N	dashed	SET NULL
41	trip_alerts	trip_id	trips	1 : 0..N	dashed	CASCADE